Laser optics for material processing

- Laser Optics Primer — synthetic demo course
- Synthetic course written for testing and demonstration.

How a laser works

Chapter

Stimulated emission and population inversion

- A photon of energy E2 – E1 can stimulate an excited atom to emit an identical photon (same frequency, phase, direction).
- Gain requires more atoms in the upper level than in the lower level: a population inversion.

[Figure] Energy-level diagram: pump → upper level → lasing transition → lower level.

Optical resonator

- Two mirrors face each other around the gain medium; one is partly transmitting (output coupler).
- Only light travelling along the axis passes the gain medium many times and is amplified.

Pumping methods

- Optical pumping: flash lamps or laser diodes excite the gain medium.
- Electrical pumping: a gas discharge (CO■, He-Ne) or a current through a semiconductor junction.

Curved resonator mirrors

- Plane-parallel mirrors are very sensitive to tilt; light walks off the axis.
- *Stability condition: $0 \le g1 \cdot g2 \le 1$ with $g = 1 - L/R$.*

[Figure] Confocal resonator: R1 = R2 = L.

Properties of laser light

Chapter

Four key properties of laser light

- Monochromatic: a very narrow range of wavelengths.
- Directional: a small divergence angle.
- Bright: high power per area and solid angle.
- Coherent: a fixed phase relation in time and space.

Power density: laser vs lamp

- *Power density I = P / A (W/m²).*

Example: A 1 W laser focused to a 100 µm spot gives about 1.3×10■ W/m²; a 100 W lamp spreads its light over the whole room.

Spectral linewidth

- Linewidth $\Delta\lambda$ (or $\Delta\nu$) measures how monochromatic a source is.
- *A narrow linewidth gives a long coherence length: $L_c \approx \lambda^2/\Delta\lambda$.*

Coherence

Chapter

Amplitude, wavelength and phase

- $E(x, t) = E0 \cos(kx - \omega t + \phi)$, $k = 2\pi/\lambda$, $\omega = 2\pi\nu$.
- Two waves interfere constructively when their phases agree.

Temporal vs spatial coherence

- Temporal coherence: phase correlation along the beam, set by the linewidth (coherence length).
- Spatial coherence: phase correlation across the beam, set by the source size.

Coherence length

- $L_c = c/\Delta\nu \approx \lambda^2/\Delta\lambda$.

Example: He-Ne laser, $\Delta\nu \approx 1.5$ GHz $\rightarrow$ $L_c \approx 0.2$ m; white light $L_c \approx 1$ µm.

- Interference fringes vanish when the path difference exceeds the coherence length.

Measuring coherence with an interferometer

[Figure] Mach–Zehnder interferometer: split, delay one arm, recombine.

- Fringe visibility drops as the delay approaches the coherence length.

Spatial coherence width

- *Coherence width $w \approx \lambda \cdot D/s$ for a source of size s seen from distance D.*

Improving coherence by filtering

- A narrow band-pass filter lengthens the coherence length; a pinhole increases spatial coherence.
- Both throw away light, so intensity drops.

Laser output: pulses and beam modes

Chapter

CW vs pulsed output

- CW lasers emit constant power; pulsed lasers concentrate energy into short pulses with high peak power.

Pulse duration and material removal

- Long (µs–ns) pulses melt and vaporise; ultrashort (ps–fs) pulses ablate before heat spreads, leaving little heat-affected zone.

Q-switching principle

- The resonator losses are kept high while the gain medium stores energy, then suddenly switched low.
- The stored energy leaves in one short pulse (ns) with high peak power.

Q-switching methods

- Active: acousto-optic or electro-optic modulators switched by a signal.
- Passive: a saturable absorber that becomes transparent at high intensity.

Transverse modes and the Gaussian beam

- TEM■■ is the fundamental transverse mode: a single bright spot with a Gaussian profile.
- *$I(r) = I0 \cdot \exp(-2r^2/w^2)$; w beam radius at $1/e^2$ intensity.*
- *Far-field half-angle divergence $\theta \approx \lambda/(\pi \cdot w0)$.*

Choosing a laser source

Chapter

Lasers by application

[Table] Cutting/welding: CO■, fibre. Marking: fibre, DPSS. Micromachining: ultrafast, excimer.

- Material absorption at the laser wavelength decides the choice.

Laser types and wavelengths

[Table] CO■ 10.6 µm; Nd:YAG and Yb fibre ≈ 1.06 µm; frequency-doubled 532 nm; excimer 193–351 nm.

Specifying a laser source

[Table] Wavelength, average power, pulse energy and duration, repetition rate, beam quality M^2, pump type.

Example: Example data sheet: flash-lamp Nd:YAG, 1064 nm, 10 ns, 20 Hz, 500 mJ.

Beam expansion and collimation

Chapter

Keplerian beam expander

[Figure] Two positive lenses separated by f1 + f2; internal focus between them.

Galilean beam expander

[Figure] A negative lens followed by a positive lens, separated by f2 – |f1|; no internal focus.

Expansion ratio and output diameter

- *Magnification M = f2/|f1|; output diameter D_out = M·D_in.*
- *Divergence is reduced by the same factor: θ_out = θ_in / M.*

Beam expander as collimator

- *Focused spot diameter $d \approx 4\lambda f/(\pi D)$: a larger input beam D gives a smaller spot.*

Beam shaping

Chapter

Purpose of beam shaping

- Many processes need uniform energy across the spot (e.g. annealing, ablation of large areas).

Beam-shaping methods and components

- Refractive shapers (aspheric lenses), diffractive optical elements (DOEs), micro-lens array homogenisers.

Top-hat vs Gaussian profiles

[Figure] Gaussian: peaked centre, soft edges. Top-hat: flat centre, steep edges.

Beam splitting and dichroic optics

Chapter

Beam-splitting methods

- Partially reflecting plates or cubes split intensity; polarising cubes split by polarisation; dichroics split by wavelength.

Dichroic mirrors

- Thin-film interference coatings reflect one wavelength band and transmit another.

[Figure] Transmission curve with a sharp edge at the cut-off wavelength.

- Example slide value: a 45° dichroic reflects 1064 nm and transmits 532 nm at 99 %.

Dichroic mirror applications

- Combining a visible pilot beam with an infrared process beam; separating harmonics after frequency doubling.

Beam scanning and delivery

Chapter

Beam-scanning devices

- Galvanometer mirrors, polygon scanners and XY stages move the spot over the workpiece.

Galvo scanner with F-theta lens

- *F-theta lens: spot position $x = f \cdot \theta$ (linear in scan angle).*
- It keeps the focus on a flat field.

Example: Measured spot 78 µm vs designed 75.4 µm: a 5.5 % deviation.

Free-space vs fibre delivery

- Free space: mirrors and lenses, any wavelength, alignment needed. Fibre: flexible, robot-friendly, limited by wavelength and peak power.

Fibre modes and single-mode output

- *Single-mode condition: $V = (2\pi a/\lambda)\cdot NA < 2.405$.*
- Multimode fibres carry more power but degrade beam quality.